

Password Strength & Breach Checker

Rule Documentation

1. Password Strength Analysis

The system evaluates password strength using zxcvbn scoring and entropy-based analysis.

2. Entropy Analysis

Entropy is estimated from password length and character-set size.

3. HIBP Password Breach Check

The password is hashed locally using SHA-1. Only the first five hash characters are sent to the HIBP Pwned Passwords range API. Returned suffixes are compared locally.

4. Email Breach Check

The application supports the HIBP v3 breached-account API when a valid API key is configured.

5. Secure Password Generator

Python's secrets module is used to generate configurable random passwords and memorable passphrases.

6. SQLite Storage

Security events are stored with timestamp, event type, severity and message.

7. Watchdog

Watchdog monitors the security log for new or modified entries.

8. Privacy

Plaintext passwords are never stored. For password breach checks, only five SHA-1 hash characters are transmitted to HIBP.

9. Limitations

This is an educational cybersecurity prototype. Live breach checks depend on HIBP availability and email checks require appropriate API access.